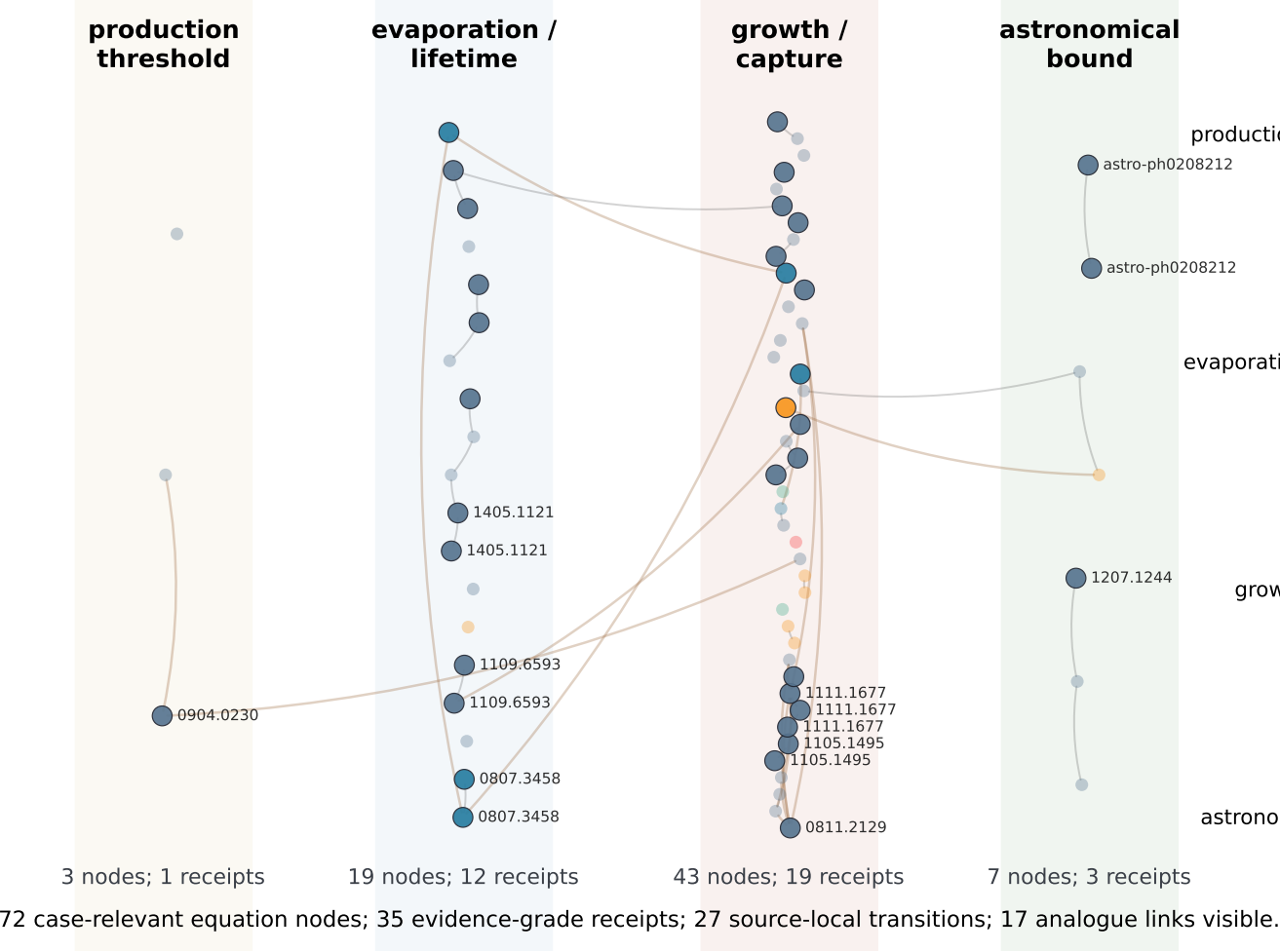
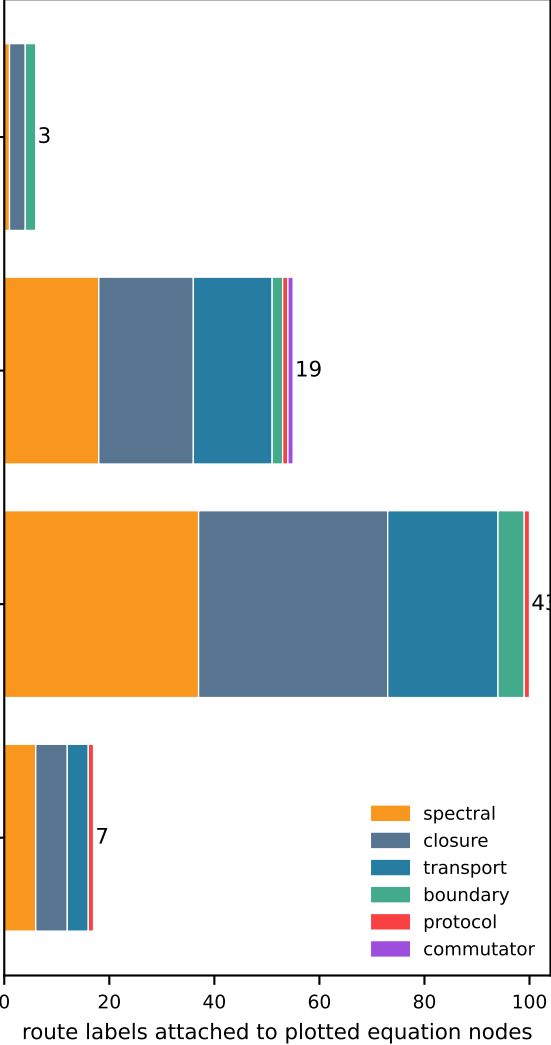


**A. Equation mechanism graph: case-relevant nodes and retained receipts**



**B. Route composition**



Interpretation: the graph is dense in spectral/closure/transport roles, but the collider side has only a production hook; survival, capture and growth are supplied as transfer constraints, not as a direct collider derivation.